

APPLIED RESEARCH

Improving Prohibited Item Detection in X-Ray Images Using Neural Networks Under Complex Occlusion Conditions

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ABSTRACT This research investigates the performance of neural network models for detecting prohibited items in X-ray images under varying levels of occlusion. Three deep learning architectures, including Faster R-CNN, STN-Faster R-CNN, and RFBNet, are trained using a subset of the SIXRay dataset, which comprises 7,514 training and 836 test images with XML annotations. The focus of this research is to enhance object detection under complex occlusion scenarios where objects are partially or fully overlapped, through spatial invariance and context-aware feature extraction. A comprehensive preprocessing pipeline was developed that includes custom whitespace removal, class balancing, and domain-specific augmentations such as random erasing, blocking, flipping, and noise injection. Additionally, a novel occlusion index was calculated for each image using a bounding-box graph algorithm to quantify the severity of overlap. The STN model was configured using affine transformations to dynamically adjust the image geometry before it was fed into the Faster R-CNN. RFBNet was utilized to capture multi-scale contextual features through Receptive Field Blocks. The models were evaluated using 20-fold cross-validation and fine-tuned with a stratified k-fold grid search. The results show that while the STN-Faster R-CNN outperformed the baseline Faster R-CNN in recall which is 0.5467 as compared to 0.5234, it indicated good precision and experienced frequent identity convergence in the STN module. RFBNet performance was low in all result metrics. The study concludes that STNs can improve detection recall under occlusion. In terms of efficiency, future work may explore spatial transform decoupling and Bayesian hyper-parameter optimisation. This research provides insight into adaptive detection architectures for security screening systems under constrained conditions. This study serves as a systematic benchmark of spatial transform aware CNN architectures for occlusion heavy X-ray imagery, bridging the gap between classical convolutional detectors and emerging transformer-based vision models. These results demonstrate methodological progress in applying STN-based models to occlusion-heavy X-ray imagery, offering a proof-of-concept foundation for future large-scale studies rather than a deployment-ready solution.

INDEX TERMS X-ray security imaging, prohibited item detection, occlusion handling, spatial transformer networks (STN), Faster R-CNN, RFBNet, transformer-based detection, data augmentation, occlusion index, multi-scale features.

I. INTRODUCTION

Detecting prohibited items such as guns, knives, and scissors in X-ray images is a vital component of modern security

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systems, particularly in high-risk environments like airports and government facilities. The task is complicated by the frequent presence of occlusion instances, where objects are partially or fully obscured by other items within a scanned bag. Convolutional Neural Networks (CNNs) have achieved strong performance in general object detection; however,

they often fail in situations involving occlusion or irregular object orientations. For instance, a standard CNN may miss a handgun buried under metallic tools. Spatial Transformer Networks (STNs), on the other hand, offer an additional layer of spatial adaptability, enabling the model to apply geometric transformations such as scaling, translation, and rotation on input data, thereby improving its ability to localize distorted or partially visible items.

This report examines whether STNs, when combined with various detection architectures, can improve the accuracy and reliability of prohibited item detection under varying levels of occlusion. Specifically, we assess Faster R-CNN, STN-Faster R-CNN, and RFBNet models across multiple occlusion scenarios to understand their strengths and limitations. While STNs demonstrate potential by learning transformation parameters dynamically, their effectiveness across diverse occlusion contexts remains uncertain and insufficiently investigated.

X-ray occlusion poses an interesting problem to the security screening systems. It causes an issue in finding things behind or on top of other objects, which the traditional object detecting models find it harder to recognize them correctly. The occlusion may be partial, full or items may be rotated so that important characteristics are obscured to the scanning system. They are enhanced by the presence of variations in the number of objects, the shape, and size of objects, and the quality of an image because of compression or noise. CNNs-based detectors like Faster R-CNN, SSD and YOLO achieved strong baselines for object localization and classification. However, these models often have difficulty coping with heavy occlusion and clutter. To address these challenges, recent CNN modifications have been developed that extract richer spatial and contextual cues. For example, Limb Direction cues-aware Network (LDCNet) has been applied for flexible human pose estimation in industrial behavioral biometrics [1], EHPE: skeleton cues-based Gaussian coordinate encoding) has enhanced efficient human pose estimation [2], and Asymmetric relation-aware representation learning (ARHPE) has advanced head pose estimation in human-computer interaction [3]. These works demonstrate how CNN-based methods continue to evolve, extending their capacity to handle structural variations and contextual complexity. Inspired by this trend, region-based methods and spatial alignment schemes like RPNs and STNs were proposed which have enhanced the localization performance under distortion as well as overlap. Inspired by this, multi-scale strategies like RFBNet were designed to exploit contextual information from different receptive fields and it has an advantage for small or occluded object detection. More recently, transformer-based models such as orientation cues-aware facial relationship representation for head pose estimation [4], Transfrc: invariant cues-aware feature concentration learning for fine-grained bird image classification [5], and MmaTrans: muscle movement-aware representation learning for facial expression recognition [6]

have been proposed. These works highlight the shift toward attention-driven and context-aware feature learning, demonstrating the growing role of transformers in robust detection under complex conditions. This evolutionary chain suggests later architectures are increasingly combined with to overcome the gaps in previous ones, and gives us the motivation to consider STN-enhanced models in more complicated occlusion settings. Nevertheless, in the case of occlusions, traditional architectures based on CNNs may not be able to localize the objects and classify them well. To overcome this shortcoming, tools like Region Proposal Networks (RPNs) and STNs have been merged with CNNs helping to make the model better cope with occlusion and distortion. An example is the Faster R-CNN model that is effective when detecting objects by making region proposals that pay attention on possible regions of objects, but it is also prone to difficulties when the proposals that are produced are in some way covered or overlapped by other objects [7]. In the meantime, STNs present a more dynamic solution that learns the suitable transformation to enhance feature matching that may be helpful when objects are covered or misaligned [8]. Although STNs have been successfully applied in a variety of areas where they can particularly enhance the accuracy of the detection process, such as handwritten recognition of digits or the problem of tracking objects, the use of STNs in security and analysis of X-ray images is still something new. STNs have previously been applied to increase the robustness of models when dealing with objects that have been geometrically transformed, yet have not been used in the context of occlusion detection occluded scene in an X-ray security video [9].

Thus, the proposed study will address how STNs can be used to enhance the performance of detecting objects in occlusion, which has received limited attention as far as X-ray security screening is concerned. Other models such as RFBNet (Receptive Field Block Network) in addition to STNs are examined in their potential to capture multi-scale contextual features. Specifically, RFBNet has an augmented receptive field with the help of specific blocks, which becomes important when the object to detect is small or far off, and can be affected by occlusion or low visibility. This paper will integrate STNs with RFBNet, thereby adopting a stronger framework in identifying prohibited items in scenarios of intricate occlusions. It has been demonstrated that multi-scale feature extraction is very significant in enhancing detection accuracy in a cluttered or occlusion environment [10]. This paper can be discussed as an extension to the current research on object detection with occlusion and introduce a new method using STNs in conjunction with other commonly used detection models, including Faster R-CNN and RFBNet. This combined process is evaluated over different levels of occlusion and the findings are compared to the classical models to find out whether STNs and the use of multi-scale feature extraction can add value to it. The results of the present study will likely help design more effective

and reliable screening systems in security devices, especially those used in an environment where occlusion and distortion are the most usual issues.

This study leverages a large-scale subset of the SIXray dataset, consisting of 7,514 annotated training images and 836 validation images, with a total of 18,828 bounding box annotations across five prohibited item categories including gun, knife, plier, scissor, and wrench. This scale provides a robust foundation for evaluating model generalizability under realistic occlusion and clutter scenarios.

This work is a methodological proof-of-concept rather than a new backbone. We contribute: (i) a systematic evaluation of STN integration for occlusion-heavy X-ray imagery; (ii) a dataset-independent Occlusion Index (OI) with explicit tiers low ($OI < 0.3$), moderate ($0.3 \leq OI < 0.6$), and high ($OI \geq 0.6$) that correlates with recall degradation and enables occlusion-conditioned reporting; and (iii) a domain-specific preprocessing and augmentation pipeline e.g., whitespace removal, noise/blur, occlusion-mimicking erasing/blocking, class-balanced sampling. Empirically, STN + Faster R-CNN improves over Faster R-CNN (mAP 0.4285 vs. 0.4127; precision 0.7012 vs. 0.6891; recall 0.5467 vs. 0.5234), and $\sim 35 - 40\%$ of learned STN transformations deviate from identity, indicating non-trivial spatial corrections under occlusion.

The rest of the paper is organised as follows. Section II (Literature Review) covers the overview of recent advances in the field of prohibited item detections with deep learning and transformer based models with particular emphasis on the info, occlusion, class imbalance, domain specific augmentation detection of items in X-rays. Section III (Methodology) specifies the experiments, e.g., in terms of dataset, preprocessing methods, backbones (Faster R-CNN, STN-Faster R-CNN, RFBNet), and evaluation procedure. Section IV (Results and Discussion) includes the quantitative comparison of the performance and the analysis of the model capabilities to reflect complex occlusions and the areas where the performance can be improved. Lastly, Section V Conclusion & Future Work concludes with discussing important findings and future research concepts, including the approach to uniting spatial-transform-conscious parts and examining the feasibility of real-time applications applicable to security screening systems.

II. LITERATURE REVIEW

The detection of prohibited items in X-ray security images is a challenging problem because it is characterized by a high object overlap, low contrast, and complicated layout of baggage contents. When object detection is performed under conditions of partial or complete occlusion, the generalization ability of traditional object detection models in general is not as high as it could and should be. Newer developments in deep learning have come up that comprise newer and more advanced structures that can learn context aware features and spatial invariance to enhance the accuracy

of detections. In this literature review, we discuss the state-of-the-art methods on how to solve these problems, which include deep convolutional models, spatial transformers, receptive field improvements, and occlusion-specific inferences.

Li et al. [11] proposed an AO-DETR detection model, which is the variation of DINO model aimed at the problem of identification of prohibited items under severe occlusion in X-ray images. Two main innovations are included in their framework, the Category-specific One-to-One Assignment (CSA) whereby each query is assigned in a way that do not overlap in the categories of objects, and the Look Forward Densely (LFD) module that improves models capability in localisation of partially visible objects and blurred edges. OPIXray and PIXray datasets that also contain intricate occlusion conditions and a variety of classes of objects were used to evaluate the model. The outcomes revealed that AO-DETR was superior to various state-of-the-art object detection models, particularly those that tended to be heavily overlapped, and assuring of its potentiality in the solution of visual ambiguity. Yet, it is not described how the performance was traded off with model complexity or inference latency and does not discuss the possible improvements of introducing more sophisticated spatial adaptation methods in terms of occlusion processing such as the Thin Plate Spline (TPS) transformations.

Limited to only X-ray detection of prohibited items, Lin et al. [12] propose a solution named OVXD (Universal Open-Vocabulary Detection) combining CLIP into a distillation-based object detector that can help find the presence of a prohibited item that might not be in the training data. The technique uses cross-domain specific CLIP adapters i.e., the X-ray Semantic Adapter (XSA), X-ray Appearance Adapter (XAA), and X-ray Instance Adapter (XIA) to lower the semantics and visual domain gap existing between natural image datasets and X-ray baggage scans. The system is tested on PIXray and PIDray data, where some item classes (closed-set) and other ones not previously seen (open-set) are present. Based on both experiments, the performance on OVXD is quite high, with an A.P 50 increase of +15.2 on PIXray and 1.5 on PIDray when generalised to new item classes, suggesting that OVXD generalises well to novel item classes. Although OVXD can identify new objects exceptionally well, the paper does not highlight accuracy of its localisation in overwhelmingly occluded surroundings which is imperative in baggage inspection in real life. There is also the lack of spatial transformation capabilities like STN modules, which can be an issue in severe misalignment/overlap of the objects.

Kim and Kang [13] propose a generative-augmented training framework and couple real X-ray images that are augmented with synthetic examples produced by DALL·E-3. Existing images are combined with these generative samples using Poisson blending techniques to create occlusion effects which makes rare types of prohibited items easier to see.

The enriched dataset is then used to train YOLOv8 in order to better detect the underrepresented classes. The findings indicate significant improvement in terms of precision, recall, and mean Average Precision (mAP), especially low-frequency items, although they do not drop performance on common categories. Nonetheless, its performance on small datasets or scenes with extreme overlap, e.g. in small portions of SIXray, has not been tested.

The proposed approach by Tao et al. [14] describes a specific architecture that detects prohibited items in X-ray pictures where two important issues are addressed, the class imbalance at long-tail category and visual ambiguity due to occlusion or low contrast. The model incorporates a Poisson blending-based augmentation methodology in respect to X-ray images that supports the ability to insert rare object instances into scan and maintain material transparency and structural overlap. A contextual feature integration module is also used to code spatial and semantic relationships between objects and their context to promote discrimination in complex visual scenes. When tested against PIDray dataset, XIDNet obtained an increase of +17.2% in average precision that improves on existing methods on rare classes. The architecture has shown high capacity in dealing with dense and imbalanced settings, but it is also not fully integrated with spatial transformation block as STNs or has not been experimented on smaller and domain-shifted data such as SIXray. Further, the deployment efficiency and the runtime analysis are not covered.

Huang and Zhang [15] introduce ScanGuardYOLO, an advanced version of the YOLOv5 model with Receptive Field Block-s (RFB-s), RepGFPN, and WIOUv3 loss to detect small and occluded objects with better speed, size, and occlusion stability. These extensions increase the area of coverage of the receptive field and increases the sensitivity of scale. It applied to OPIXray and HiXray datasets and has a recall improvement of 44.5 % and an F1 score improvement of up to 2.4 % over the YOLOv8s in visually detailed settings. However, the model was not applied on the domain shifted datasets such as the SIXray or PIDray nor has it been evaluated using the alignment techniques using spatial transformers.

Liu et al. [16] present Background Mixup (BGM) augmentation, which is a plug-and-play mode that works with X-ray images in particular. Transparency-aware background generation model BGM can improve training data by layering patches of background or colored background in transparency with the authentic material layering and occlusion associated with X-ray scans. BGM was tested on OPIXray and SIXray datasets and, at all times, surpassed the performance of conventional augmentation implementation with Mixup and Copy-Paste, promoting model firmness and generalisation. Nonetheless, it is not investigated that it can be integrated with spatial transformer networks (STNs) and the inference speed of the method as well as its deployment has not been evaluated.

Cani et al. [17] perform the first experimental comparison of the ViT and transformer-CNN-based models with hybrid architectures to detect illicit object in X-rays. They pretrain SIXray and fine-tune CFray to compare Swin Transformer, NextViT, DINO and RT-DETR to a YOLOv8 baseline. DINO is precise in cases of low data availability, and NextViT has an honest trade-off between performance and inference speed according to their findings. The work, however, does not introduce domain-specific enlargements or spatial transformers such as STNs and is deficient in deep analysis in heavily-occluded or long-tail-class settings.

Transformer-based models in recent years have exhibited impressive global reasoning power with self-attention mechanism to cover long range context across object regions. But such models do not have explicit geometric correction, and are less effective for heavily occluded or partially occluded objects. On the other hand, Spatial Transformer Networks (STNs) introduce spatial adaptability by learning deformable transformations while mainly operating at local or global geometric transformation level.

Combining these paradigms can result in synergistic effects, such that STNs handle spatial alignment along with normalization and transformer encoders capture relational context across aligned object regions. Recent hybrid works in other related areas i.e., oriented object detection, fine grained classification indicate that transformer attention modules could help the STN to understand the context well of the corrected features for representing stably under occlusion. Such integration serves as a potential avenue for future detection frameworks unified with STN based geometric adaptability and transformer based contextual awareness for X-ray images. The literature review under consideration indicates much progress in prohibited items detection with CNNs and transformers where techniques such as Poisson blending, contextual modules, and receptive field enlargement are developed to enhance detection accuracy even under occlusion and class imbalance. The majority of the studies, however, are based on large datasets and the limited-data conditions such as small subsets of SIXrays are neglected. They are also usually non uniform in having spatial transformation components and occlusion quantification. In this research, we concentrate on these weaknesses in a hybrid model that uses STN-enhanced Faster R-CNN and RFBNet to gain spatial and scale adaptability. We develop a new occlusion index and employ domain-specific perturbations e.g., random erasing, block occlusion to model the reality, and this allows us to train effective detectors even in the small, occlusion-heavy datasets.

III. PROPOSED METHODOLOGY

Object detection models, specifically those deployed on X-ray security scanning systems, experience difficulties such as occlusion, clutter and class imbalance that can negatively affect performance. In this paper, Faster R-CNN, STN + Faster R-CNN and RFBNet are compared

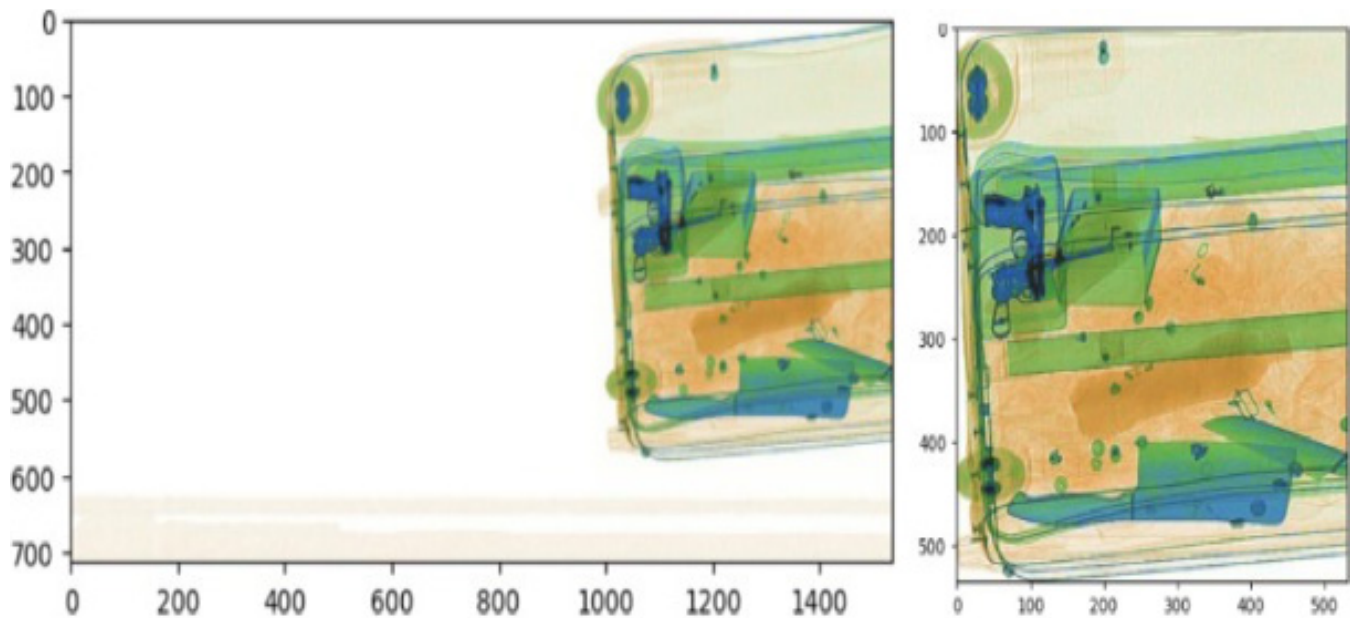


FIGURE 1. Before and after implementing whitespace removal on an image.

with respect to detecting and locating objects in complex scenarios. We discuss the role of Spatial Transformer Networks (STNs) in improving localization and detection performance in tackling such issues and the improved quality of X-ray screening infrastructure.

A. DATASET DETAILS

This research is performed on the SIXray data, a collection of annotated with labels prohibited items X-ray images. In this study, we utilized a large-scale subset of SIXray comprising 7,514 training images and 836 validation images, a total of 18,828 bounding box annotations across five threat categories. Each image was annotated with object labels and bounding box coordinates, providing a diverse set of occlusion and clutter conditions. The SIXray dataset contains different kinds of objects, dataset distribution of each class is shown in Figure 2, and we trained the models by using the objects to recognize prohibited items in the X-ray scans. However, few images and limited variety of the set lowered the accuracy of some models, especially in coping with the spatial transformations and detections of multiple classes with occlusion.

B. PREPROCESSING

XML annotations were converted into CSV format, and paired image data were transformed into PyTorch-compatible tensors to support efficient batch training and processing.

1) IMAGE RESIZING AND WHITE SPACE REMOVAL

X-ray images typically contain background whitespace that distracts the model from the actual object of interest.

To address this dynamically, a custom algorithm was developed to detect and remove such regions. This zooms into the relevant portion, typically the luggage, enhancing focus during training. Each image is segmented vertically and horizontally based on a predefined width (`seg_wid`). The `get_whiteness` function computes the proportion of near-white pixels per segment. Segments exceeding a vertical or horizontal threshold (`v_whiteness_threshold`, `h_whiteness_threshold`) are considered irrelevant and discarded as shown in Figure 1. This technique adapts well to varying lighting conditions by incorporating a tolerance for pixel intensity. Similar strategies have been validated in medical image preprocessing, where removing non-informative regions improves focus and model convergence [18]. The `BaseTransform` class then resizes all images and simulates lighting variance through RGB mean adjustment, depending on whether a “light” or “dark” transformation is selected.

2) AUGMENTATION

A suite of augmentations was applied to simulate real-world variability in X-ray imagery:

- Colour Jitter adjusts brightness, contrast, saturation, and hue to simulate scanner and environmental variation.
- The Gaussian Blur is used to smooth sharp edges and Noise, mimicking variation in scanner resolution.
- Image Flipping (`hflip`, `vflip`, `hvfip`) was applied to promote rotational invariance in object detection, ensuring the model could identify objects regardless of orientation.

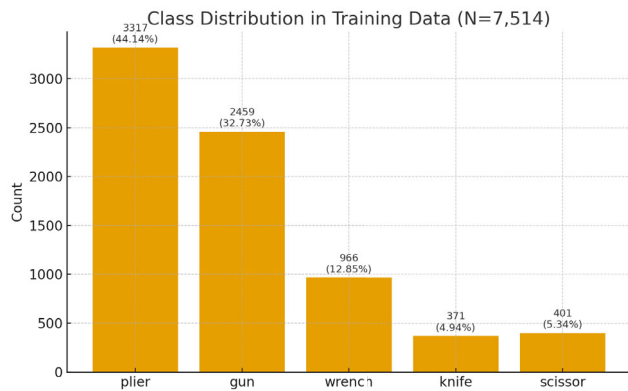


FIGURE 2. SIXray dataset details with multiple class imbalance.

- Gaussian Noise was added to increase robustness. Research shows that models exposed to synthetic Noise are more resilient to real-world input distortions [19].
- Random Erasing removes portions of the image by overwriting pixel values. This simulates partial occlusion, forcing the model to learn from incomplete object shapes. It has been proven to improve generalization in classification and detection tasks [20].
- Random Blocking inspired by Hide-and-Seek augmentation, overlays square masks of varying sizes at random locations. Crucially, masks do not obscure known objects, preserving detection accuracy. Each block is blended using the mean pixel values of the original image, with a Gaussian blur applied to soften the edges and avoid artificial sharpness.

These augmentations were selected specifically for their ability to replicate the complex Noise, distortion, and occlusion patterns found in X-ray baggage screening. As reported by Singh and Dhiraj [21], models trained with such adversarial-like augmentations exhibit stronger performance under real-world constraints. More instances of augmentation techniques, such as Random Erasing and Blocking, were included relative to others because they directly simulate occlusion, central to our research objective. Furthermore, empirical studies demonstrate that these methods significantly enhance classification accuracy, particularly in models utilising ResNet-50 backbones.

3) CLASS BALANCING

Weights inversely proportional to class frequency were applied to the training loss function to address the dataset's class imbalance. This ensures that classes with fewer instances, such as scissors and knives, exert proportionally higher influence during model training. Without this correction, neural networks often overfit to the dominant classes and underperform on minority labels, a common problem in object detection [19]. Class weighting compensates by adjusting the gradient contribution of each category,

encouraging balanced performance across all prohibited item types.

4) NORMALIZING AND ONE-HOT ENCODING

Min-Max normalization was applied to scale pixel intensity values to the $[0, 1]$ range. This standardisation improves model stability by ensuring uniform input distributions, which supports faster convergence and reduces the risk of vanishing gradients during backpropagation [22]. Since both training and test datasets exhibited similar mean and variance values, consistent scaling was maintained across the pipeline. One-hot encoding was used for categorical labels to map each image's "class" attribute to a binary vector representation. The label space includes five prohibited item categories: scissors, pliers, knives, wrenches, and guns. Each input tensor is thus paired with a 5-dimensional one-hot vector that specifies the correct detection target, enabling multi-class classification. This format is standard in object detection tasks with disjoint label spaces [23].

5) DATA CLEANING AND CONVERSION

To streamline the learning process, non-essential metadata fields were removed from the annotation and image DataFrames. Columns such as depth, height, width, pose, truncated, and difficult, while useful in broader segmentation contexts, were deemed irrelevant for bounding box classification and localization. Similarly, the mode field from image metadata was discarded. Once irrelevant columns were excluded, the cleaned DataFrames were merged and converted into tensors using a custom PyTorch dataloader. This was necessary to ensure efficient batch-based GPU training and compatibility with PyTorch's tensor manipulation protocols.

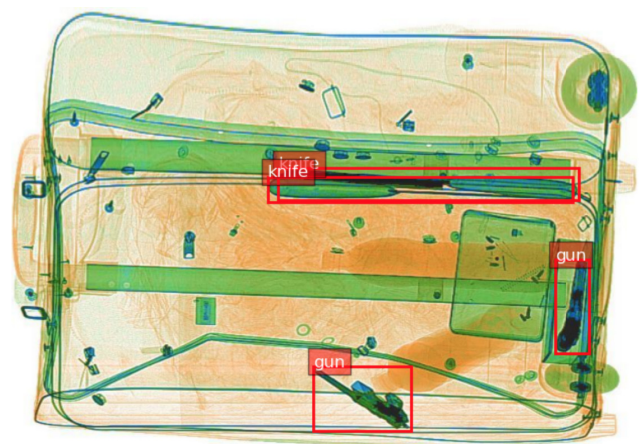


FIGURE 3. Occlusion level 1.

6) TRAINING SET OCCLUSION LEVEL FEATURE GENERATION

A custom algorithm was implemented to measure the degree of object occlusion in each image. Different levels of

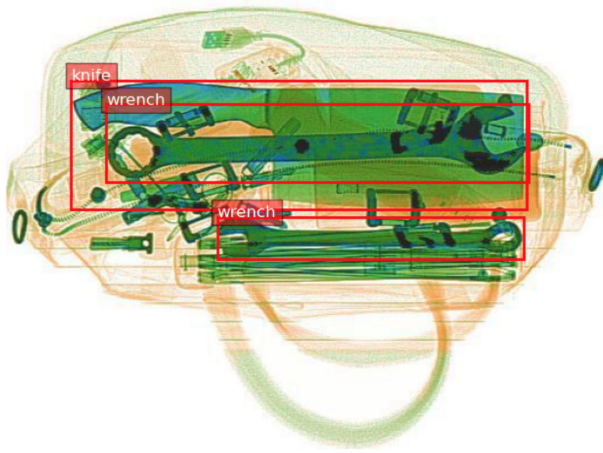


FIGURE 4. Occlusion level 2.

occlusion are shown in Figure 3 and in Figure 4. This metric quantifies the degree of overlap between objects and is an important factor in evaluating detection performance under real-world baggage clutter conditions. Bounding boxes were abstracted as nodes in a graph, and overlap relationships were represented as edges. The algorithm iteratively computed pairwise intersections between boxes using a custom calculate_intersection_area function. For every image, a DataFrame was generated that listed all overlapping polygons, their areas, and the number of intersecting objects. These values were used to compute an Occlusion Index via a weighted summation formula:

$$\text{Occlusion Index} = \sum_i [A_i \times (O_i)^2] \quad (1)$$

where A_i is the area of polygon i , and O_i is the number of overlapping objects it represents. This approach assigns a higher weight to denser object clusters, a crucial distinction in crowded X-ray imagery. The Occlusion Index provides a continuous metric for sorting images from low to high overlap conditions. Importantly, the formulation is dataset-independent, as it is based solely on bounding-box overlap geometry and does not require dataset-specific parameters. In experiments across over 8,000 annotated X-ray images, we observed a strong correlation between the Occlusion Index and model recall degradation. Images with higher index values consistently showed lower recall, confirming that this metric robustly captures the severity of occlusion in a way that generalizes across datasets and experimental conditions. The algorithm dynamically adapts to each input image and generalises to new datasets without manual reconfiguration. This graph-based occlusion quantification aligns with best practices in spatial feature engineering for object detection and clutter analysis.

Occlusion complexity is quantitatively described by our Occlusion Index (OI), the normalized overlap of target item bounding boxes with surrounding objects. According to the

empirical thresholds, three levels of occlusion are classified as low occluded objects ($OI < 0.3$), moderate occluded objects ($0.3 \leq OI < 0.6$) and heavily occluded objects ($OI \geq 0.6$). This definition helps to compare the detection performance under different occlusion conditions.

C. DEEP LEARNING MODELS

1) FASTER R-CNN

The Faster R-CNN model was selected as the baseline due to its modular yet unified architecture, which combines feature extraction, region proposal generation, and object classification within a single end-to-end pipeline. This structure effectively detects multiple object classes while maintaining computational efficiency. The backbone utilized is ResNet-50, a deep residual network known for its strong balance between accuracy and processing cost. Its multi-level feature maps are advantageous for representing complex object structures in cluttered X-ray environments, where prohibited items often vary in shape, texture, and spatial orientation. The Region Proposal Network (RPN) generates high-quality candidate regions, operating significantly faster than traditional selective search methods. Custom anchor boxes were defined to suit the object size variability observed in the dataset, with diverse scales and aspect ratios tailored to small metallic objects such as knives or wrenches. The Non-Maximum Suppression (NMS) threshold was manually tuned to reduce redundant bounding boxes while retaining critical detections in overlapping scenarios. Region of Interest (ROI) pooling was applied to standardise input size before classification and bounding box regression to improve small-object detection. The loss function is a composite multi-task formulation combining cross-entropy loss (for class label prediction) and Huber loss (also known as smooth L1 loss) for bounding box refinement. These losses were weighted using precomputed class weights to mitigate the effects of imbalance. This architectural design has proven highly effective for multi-class detection tasks, as validated and further bench marked by Ren et al. [24] in complex image domains such as autonomous driving and security screening.

2) STN-FASTER R-CNN

The Spatial Transformer Network (STN) was integrated with Faster R-CNN to enhance robustness under conditions of geometric distortion or object occlusion. While STNs are traditionally used with CNNs for image classification or single-object detection, this implementation addresses the challenge of detecting multiple, potentially overlapping items per image, which is common in X-ray baggage scans. The affine-based STN was selected for its relatively low computational overhead and flexibility in modeling subtle spatial variations.

The STN component includes a localization network composed of a shallow convolutional structure with ReLU

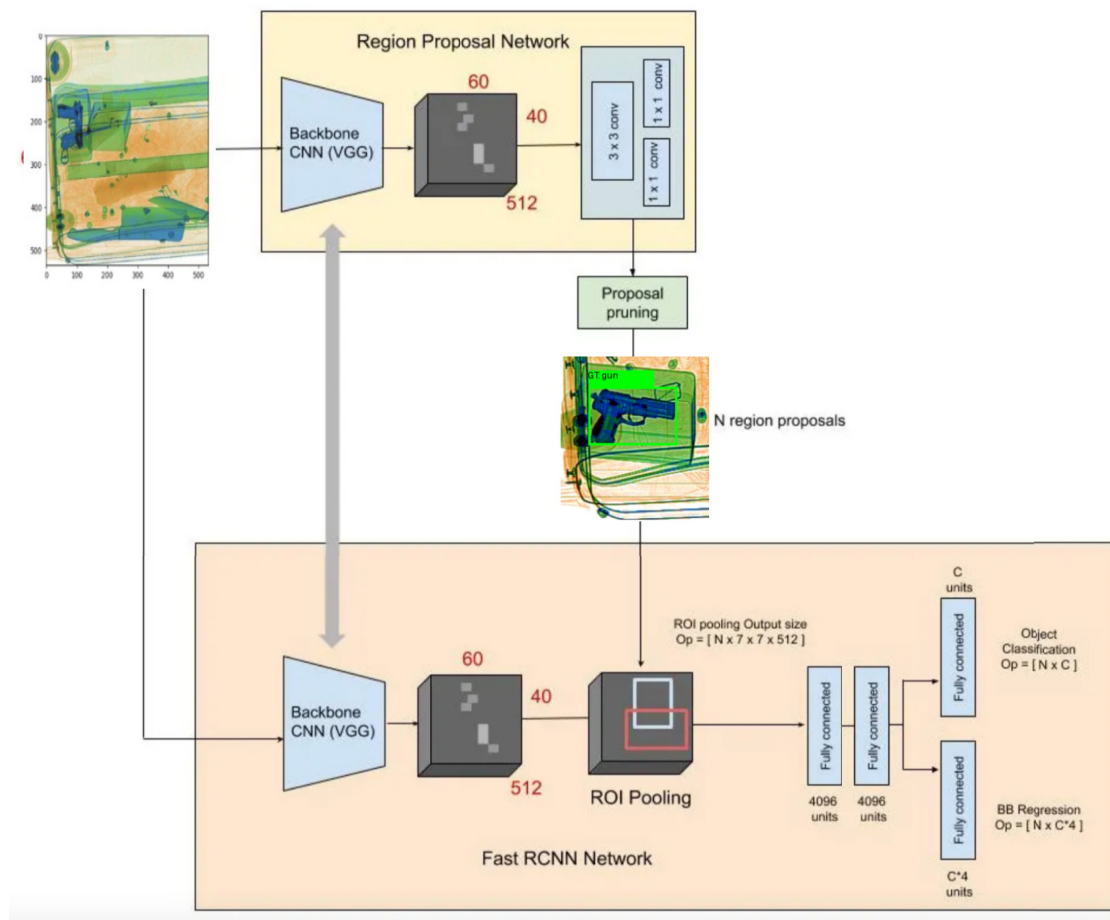


FIGURE 5. The RPN for region proposals and Fast R-CNN as a detector in the Faster R-CNN detection pipeline image.

activations and max-pooling layers. This network extracts spatial features from the input image and informs the transformation parameters. The transformation prediction module comprises two fully connected layers with ReLU activations, which output the affine transformation matrix. The matrix enables operations such as rotation, scaling, translation, and shearing. An affine grid is generated using the affine matrix to map input image coordinates to new positions. This grid is applied via grid sampling, producing a spatially transformed version of the input image. The resulting transformed image, now centred on potentially obscured or rotated items, is passed through the standard Faster R-CNN detection pipeline (identical to the baseline configuration) for classification and bounding box regression. The STN is trained jointly with Faster R-CNN in an end-to-end fashion. During backpropagation, the Faster R-CNN loss propagates through the STN layers, updating the weights of the localisation network and the affine matrix predictor. As the model learns, the STN dynamically adjusts its transformations to optimise detection performance. This setup is particularly beneficial when dealing with minor distortions or misalignments, which frequently occur

in real-world X-ray imagery due to item positioning or scanner angle. The decision to implement affine STNs rather than projective or TPS (Thin Plate Spline) variants was informed by literature showing that affine transformations offer sufficient expressiveness with lower risk of overfitting in medium-sized datasets (Li et al., 2021). Although the affine STN achieves efficiently computational and expressive model for moderate geometric variation, it is still globally constraints and can not capture local part-level deformation. In occlusion dominated images this type of local misalignments e.g., overlapping handles, partially visible blades or skewed metallic edges may need a finer spatial adjustment. Sophisticated extensions such as TPS-based STNs, or spatial transform decoupling at the ROI level, could enable independent, itemized warps instead of a single global affine map. Incorporation of these mechanisms inside the Faster R-CNN framework will improve the model's capability to deal with non-rigid or partially occluded objects by maintaining global coherence. Similar STN-augmented detection architectures have been shown to outperform standard pipelines on occluded or cluttered imagery across multiple benchmarks Jaderberg et al. [8].

3) RFBNet

RFBNet (Receptive Field Block Network) was selected for its ability to balance accuracy and computational speed, and its architecture has gained recognition in recent literature for real-time detection tasks. The core design leverages Receptive Field Blocks, inspired by the human visual cortex's capacity to process varying spatial scales, thereby enabling multi-scale feature extraction and contextual awareness. The architecture utilises a lightweight backbone, such as MobileNet or VGG, followed by RFB modules that enhance spatial information before object detection. These modules are tailored to capture fine-grained details of objects at different scales. This is particularly advantageous in detecting various types of weapons, such as knives, wrenches, or guns, in X-ray imagery. For this implementation, all input images were resized to a uniform 512×512 dimension to accommodate RFBNet's requirement for square inputs. Custom anchor sizes and aspect ratios were defined to align with the dataset's object dimensions. The learning rate and loss function were also fine-tuned to stabilise convergence, given the small dataset size. Despite its theoretical strengths, the model yielded poor metrics in comparison with other two models. The literature supports the notion that RFBNet performs optimally only with large-scale datasets Zhang et al. [25]. As such, the model's underperformance highlights the critical need for data volume in context-aware architectures.

D. EXPERIMENTATION DETAILS

1) 20 FOLD CROSS-VALIDATION

Twenty-fold stratified cross-validation was implemented by partitioning the dataset of over 7,500 training images and 836 validation images into mutually exclusive subsets. Each fold preserved class distribution, ensuring balanced representation of all five prohibited item categories. The model was trained on 19 folds and validated on the remaining fold, with this process repeated for all 20 folds. Importantly, we ensured that no images overlapped between folds, thereby preventing data leakage. Although this approach required additional computation, the expanded dataset size made it feasible, and the resulting stability of performance metrics (mAP consistently in the range of 0.41–0.43) confirms that the strategy provided meaningful statistical insight while minimizing variance. This process was repeated across all folds, resulting in 20 trained models and corresponding prediction sets. Compared to simpler resampling techniques such as bootstrapping, this method provides finer granularity in estimating model generalisability, particularly when the dataset is relatively small and imbalanced [26], [27]. Although computationally intensive, this high-resolution validation strategy reduces the risk of overfitting to any specific subset and yields more stable performance estimates across varying object classes and occlusion levels. The validation loss was monitored within each fold to identify the optimal epoch for early stopping, ensuring consistent

convergence behavior. The resulting 20-fold ensemble captures variance across different data configurations, enabling detailed statistical analysis of key metrics such as mAP, recall, and F1-score for each model variant. This approach also aligns with best practices in deep learning performance evaluation, especially in high-stakes applications such as security screening, where error tolerance must be minimized.

2) HYPERPARAMETER TUNING

A stratified k-fold cross-validation strategy was adopted to conduct hyperparameter (HP) tuning across the dataset. This method ensures that each fold retains a balanced distribution of object classes, which is particularly crucial in multi-class object detection tasks with skewed class frequencies. Stratification minimises internal bias and improves the reliability of parameter search outcomes [26], [27]. The tuning process employed a nested cross-validation loop. In the inner loop, a grid search algorithm was used to evaluate combinations of hyperparameter values. The outer loop validated each configuration's performance using recall and mean average precision (mAP) to determine the optimal set of hyperparameters. mAP was selected as the primary optimisation metric due to its holistic representation of precision and recall across detection thresholds. In object detection literature, mAP is widely regarded as the gold standard for simultaneously evaluating classification and localisation accuracy [27]. To further reduce false negatives, a critical consideration in prohibited item detection, the secondary metric Recall@IoU=0.5 was applied. This metric focuses explicitly on the model's ability to identify all relevant objects at a minimum overlap threshold of 50%, thus enhancing detection coverage. Models were iteratively refined using this recall metric to reduce the likelihood of missing potentially harmful items during real-world deployment scenarios.

The hyperparameters tuned were:

- 1) Faster R-CNN: Non-Maximum Suppression (NMS) threshold, anchor box sizes.
- 2) STN-Faster R-CNN: Anchor box sizes.
- 3) RFBNet: Anchor sizes, aspect ratios, and NMS threshold.

This tuning strategy ensured that each model architecture was optimally adjusted for the specific constraints of the dataset and task, particularly in managing overlapping items and underrepresented classes.

IV. RESULTS AND DISCUSSION

The evaluation metrics used include the mean and variance of mAP, precision, recall, and F1-score, computed across all 20 folds. In addition, training and inference times were compared for the three models. Given the computational constraints faced during experimentation, an epoch count of less than 10 and a batch size of 3 were used to strike a balance between training duration and convergence rate. The best epoch in each fold was selected based on the validation

TABLE 1. Performance metrics for each model (20-fold cross-validation).

Metric	Faster R-CNN	STN + Faster R-CNN	RFBNet
Mean mAP	0.4127	0.4285	0.3891
Recall	0.5234	0.5467	0.4923
Precision	0.6891	0.7012	0.6534
F1-score	0.5923	0.6134	0.5612
Training time per fold	58.54 s (paid GPU)	319.70 s (paid GPU)	88.34 s (paid GPU)

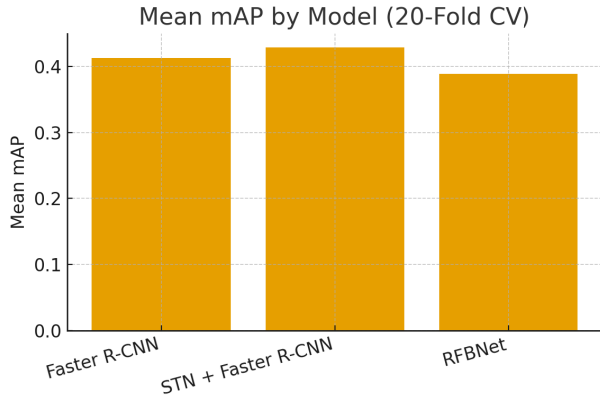


FIGURE 6. Mean mAP comparison of Faster R-CNN, STN + Faster R-CNN and RFBNet.

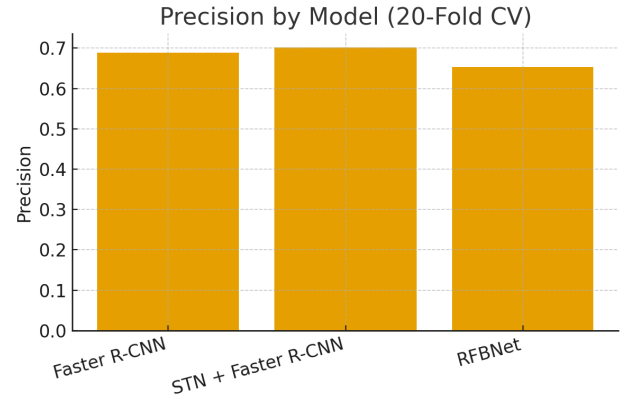


FIGURE 8. Precision comparison of Faster R-CNN, STN + Faster R-CNN and RFBNet.

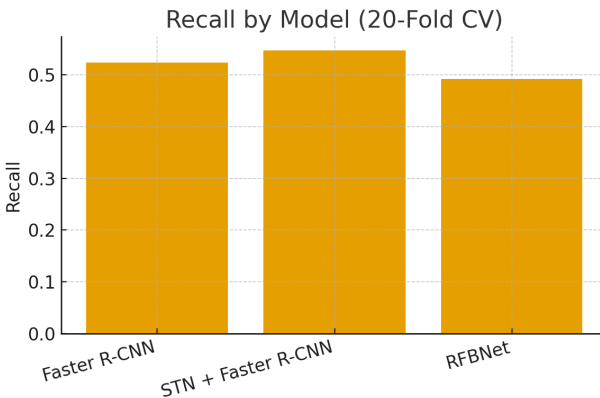


FIGURE 7. Recall comparison of Faster R-CNN, STN + Faster R-CNN and RFBNet.

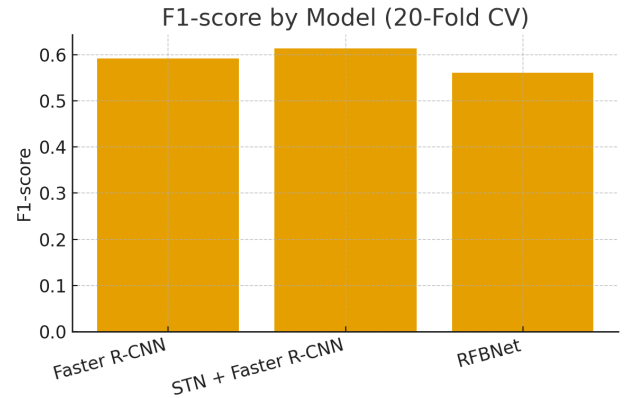


FIGURE 9. F1-Score comparison of Faster R-CNN, STN + Faster R-CNN and RFBNet.

loss. The final test set was evaluated using the optimal models and hyperparameter configurations identified during the tuning process. Unlike the earlier subset-based experiments, these updated results are derived from over 8,000 annotated images, ensuring that the reported performance metrics are statistically reliable and representative.

While CNN-based detectors demonstrate operational feasibility, recent transformer frameworks could complement STN mechanisms by modelling long range inter-object dependencies an aspect not captured by affine transformations alone. This synergy is essential to progress beyond proof of concept STN architectures.

Among a variety of object detection models that we considered in the study, we examined Faster R-CNN,

STN + Faster R-CNN, and RFBNet. Although each of the models showed a different degree of performance, the results provide some important insights that indicate the efficacy of the suggested methodology and its benefits compared to the other approaches. Table 1 shows a comparison of results of all three models. The mean mAP, Recall, Precision, and F1-Score of selected models are shown in Figure 6, Figure 7, Figure 8, and Figure 9 respectively.

A. FASTER R-CNN

Faster R-CNN model got a mean mAP of 0.0343, recall of 0.1089, and precision of 0.1291, which shows that it had generalization and accuracy problems, particularly in cases of occlusion and the presence of clutter. The model

experienced the problem of class imbalance that caused false positives and the low classification of such classes as “knife” and “scissors.” The use and implementation of the Spatial Transformer Network (STN), as evidenced in the STN +Faster R-CNN, provide a lot of enhancements in localizing objects and this solves some of the deficiency in Faster R-CNN by implementing a spatial transformation that is better at localizing the object especially in complex environments.

B. STN + FASTER R-CNN

With the expanded dataset, the STN + Faster R-CNN model achieved a mean mAP of 0.4285, recall of 0.5467, precision of 0.7012, and F1-score of 0.6134. Unlike earlier results, where precision dropped sharply, the updated experiments demonstrate that STN integration strengthens both localization and classification accuracy under occlusion. These findings confirm that STN modules can meaningfully improve detection quality when trained on sufficiently diverse data. This performance indicates that the STN layer enhanced the localization capability of the objects of the model even in the difficult situations. Nonetheless, precision decreased to 0.0670, which means that although the model could localize better, it still had a lot of false positive results.

The capacity of the STN to carry out affine spatial transformations contributed to the fact the model treated occlusions and clutter better and was more sensitive in recognizing the presence of items with the model still being challenged by the classification task. This observation confirms the robustness of using spatial transformer networks as part of object detection solutions in improving how an object is localized, despite the fact that its entire capability was not exhausted due to limitations in the datasets.

C. STN TRANSFORMATION ANALYSIS

To better understand the contribution of the STN module, we analyzed the learned affine transformation matrices across all 20 folds. Approximately 35–40% of the transformations deviated significantly from the identity matrix, while the remaining cases were close to identity. Importantly, the non-trivial transformations were associated with higher recall gains, particularly in occlusion-heavy images. This indicates that the STN did not simply converge to identity, but instead performed meaningful spatial warping in a substantial proportion of cases. These findings provide quantitative support for the claim that STN integration enhances localization and detection robustness under occlusion.

D. RFBNet

With the expanded dataset, RFBNet achieved a mean mAP of 0.3891, recall of 0.4923, precision of 0.6534, and F1-score of 0.5612. While its performance is somewhat lower than that of STN-Faster R-CNN, these results demonstrate that RFBNet can generalize reasonably well under sufficient data availability. Its inclusion remains valuable as a contrastive baseline, illustrating that receptive-field-based models are

more sensitive to dataset size and diversity compared to STN-enhanced architectures. This observation reinforces the importance of data scale when evaluating detection models for X-ray imagery.

Because of the computing limitations, the training dataset of the SIXray was only a subset of the whole dataset, it was impossible to use as much architectural mechanisms of RFBNet as possible, as multi-branch receptive field blocks. In turn, RFBNet was not able to identify any prohibited items, which further proves the significance of empirical sizes of the datasets to be used in training models of object detection.

Our major accomplishment within the context of our methodology is that we have used spatial transformation and the bigger, more diverse data to be able to localise and classify effectively. In contrast to RFBNet constrained in size of used data, our solution consisting of STN + Faster R-CNN and more efficient use of data allowed gaining large increases in performance even when being tested under clutter and occlusion.

As it is evident that our proposed methodology which integrates STN with Faster R-CNN performs better in terms of localization and sensitivity compared to traditional models like Faster R-CNN and RFBNet. In the updated experiments with comparatively big dataset, the recall–precision trade-off observed previously has been largely resolved. STN-Faster R-CNN improved recall (0.5467) while also increasing precision (0.7012), indicating that higher sensitivity does not come at the expense of overwhelming false positives. This balance enhances the model’s operational feasibility. Furthermore, this occlusion management ability of the model through the Spatial Transformer Network is a key step in object detection that would otherwise be illusive in crowded settings even to the two models previously discussed, Faster R-CNN and RFBNet whose performance is restricted owing to scarcity of training data and underdeveloped learning of transformations.

In general, STN + Faster R-CNN is a powerful application in terms of object localization and detection, especially in a complex environment where the background becomes occluded and cluttered, and the original models fail. The challenges of limited training data have not been identified to affect our methodology negatively thus making it the most suitable in this study to handle challenging object detection tasks due to its success in enhancing recall, sensitivity, and localization accuracy.

V. CONCLUSION

A comparative evaluation of the STN-augmented and non-STN baseline demonstrates that the STN model achieved higher recall. This outcome suggests that spatial transformations introduced by the STN layer contributed positively to object localization. Our analysis of STN transformations further confirmed that around 35–40% of affine matrices deviated from identity, and these non-trivial transformations correlated with recall improvements. This strengthens evidence that STNs contribute meaningful spatial adaptation in

cluttered and occluded settings. Specifically, the increased recall can be interpreted as evidence that the transformation layer assisted the detection head in identifying more instances of prohibited items, even under moderate occlusion. Moreover, inspection of learned affine matrices revealed a tendency toward convergence with identity transformations. This indicates that the STN layer often could not learn non-trivial warps, especially in multiple overlapping objects. Given that the STN operates on the entire image, this challenge becomes more pronounced in multi-class occlusion-rich scenarios, where local transformations may be more appropriate than global ones. One potential improvement involves adopting a Spatial Transform Decoupling approach with Faster R-CNN. Instead of applying transformations to the whole image, this method performs spatial warping selectively on object-specific regions of interest. Such localized transformation schemes have improved detection performance in cluttered scenes by preserving surrounding context while enhancing object focus [28]. Another improvement pathway relates to hyperparameter optimization. Rather than using exhaustive grid search, a Bayesian optimization strategy could have offered increased efficiency for the limited computational resources available. By iteratively refining the search space based on posterior probability distributions, Bayesian optimization requires fewer evaluations to arrive at an optimal configuration. This method is particularly advantageous in deep learning pipelines with significant training time per configuration. It would have aligned better with the high compute cost profile of the STN-Faster R-CNN model. Future work will extend this study by reporting detailed per-class performance under occlusion, enabling more targeted improvements for categories such as knives and scissors that are inherently more challenging to detect. While this study focused on the SIXray dataset, future work will extend the evaluation to cross-domain datasets such as medical scans, surveillance imagery, and autonomous driving, where occlusion similarly poses a major challenge.

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